

PENN VALLEY, PA, USA

WMO: 725105

Lat: 40.821N

Lon: 76.864W

Elev: 444

StdP: 14.46

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 14770

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.8	10.5	-5.8	4.1	8.6	-0.9	5.3	14.2	20.3	27.6	18.4	27.5	3.8	340	0.437

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	90.6	74.1	88.1	72.9	85.4	71.6	76.9	86.1	75.4	83.8	74.0	82.0	5.7	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.1	129.4	81.1	72.7	123.2	79.7	71.7	118.9	78.6	40.5	86.2	39.0	83.7	37.7	81.9	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	15.3	12.6		-2.0	95.2	6.0	3.5	-6.3	97.7	-9.8	99.7	-13.2	101.7	-17.5	104.2
				-2.7	80.2	5.7	2.1	-6.8	81.7	-10.1	82.9	-13.3	84.0	-17.5	85.5
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.7	28.0	30.3	39.3	50.9	61.3	69.9	73.4	71.5	64.8	53.6	42.4	33.2	
	DBStd	17.77	10.05	9.08	9.46	8.50	7.72	5.94	5.06	4.88	6.62	8.09	7.96	8.46	
	HDD50	2510	682	554	352	88	5	0	0	0	0	54	251	524	
	HDD65	5659	1146	973	796	428	168	23	4	7	86	364	678	986	
	CDD50	3121	1	1	22	116	357	597	726	667	443	165	23	3	
	CDD65	793	0	0	0	6	55	170	264	209	78	11	0	0	
	CDH74	7178	0	0	8	154	585	1534	2388	1727	691	91	0	0	
	CDH80	2347	0	0	0	44	174	494	880	542	197	16	0	0	
Wind	WSAvg	4.3	5.2	5.3	6.1	5.9	4.5	3.9	2.9	2.5	2.9	3.7	4.5	4.7	
Precipitation	PrecAvg	40.9	2.9	2.4	3.1	3.4	3.7	4.4	3.7	3.7	4.1	3.2	3.4	3.0	
	PrecMax	60.4	7.8	7.0	6.3	8.8	8.7	18.1	7.2	9.3	12.0	8.8	7.5	7.0	
	PrecMin	29.6	0.4	0.3	0.8	0.8	0.4	0.4	1.3	0.6	0.7	0.1	0.8	0.4	
	PrecStd	7.3	1.8	1.4	1.3	1.7	1.7	2.8	1.7	1.9	2.6	2.0	1.5	1.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.2	63.0	73.3	86.0	89.6	92.3	95.3	93.0	89.9	81.6	69.9	62.3	
		MCWB	55.7	54.4	57.2	64.3	70.8	74.3	75.5	74.7	72.6	69.0	59.7	56.3	
	2%	DB	52.3	53.5	66.0	78.6	84.9	88.7	90.9	88.7	85.5	75.8	64.4	54.4	
		MCWB	48.1	45.8	53.3	61.3	68.6	72.4	75.2	74.4	71.3	65.1	56.6	49.2	
	5%	DB	45.7	47.8	60.1	72.5	80.8	85.5	88.1	85.5	81.6	72.0	60.8	49.1	
		MCWB	41.0	41.5	50.6	57.8	66.5	71.2	73.8	72.7	69.1	63.3	53.5	45.2	
	10%	DB	41.9	43.3	54.7	67.0	76.3	82.3	84.9	82.5	77.9	67.8	56.5	45.5	
		MCWB	37.9	38.2	47.1	54.5	64.3	69.5	72.3	71.2	67.2	60.4	50.4	41.7	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.6	56.6	60.7	68.3	74.2	76.7	79.4	79.0	76.4	71.6	62.5	58.0
			MCDB	60.0	62.0	69.5	79.2	83.6	86.9	89.9	87.7	84.9	78.2	66.6	61.3
2%		WB	47.9	47.3	56.5	63.5	71.0	75.0	77.2	76.6	73.7	68.6	58.8	50.7	
		MCDB	51.5	53.4	63.6	73.9	81.4	84.7	87.1	84.6	81.3	74.3	63.4	53.2	
5%		WB	42.1	42.4	52.1	60.2	68.9	73.4	75.5	74.9	71.6	65.1	54.9	45.9	
		MCDB	45.1	46.6	58.9	69.8	77.6	82.3	84.2	82.2	77.7	69.9	59.1	48.7	
10%		WB	38.3	38.5	47.8	56.9	66.5	71.7	74.2	73.2	69.7	61.3	51.2	42.2	
		MCDB	41.3	43.0	53.9	64.8	73.7	79.5	82.2	80.1	74.8	65.9	55.6	45.4	
Mean Daily Temperature Range	5% DB	MDBR	15.8	17.5	19.5	22.7	22.1	20.8	21.5	21.2	22.5	20.8	18.9	14.4	
		MCDBR	20.0	24.9	28.6	32.7	27.9	23.9	24.4	24.2	26.9	26.4	25.7	18.5	
		MCWBR	16.0	18.0	17.9	17.2	13.7	10.6	10.6	11.7	13.6	15.4	17.6	15.4	
	5% WB	MCDBR	18.8	22.2	25.3	27.6	23.4	20.9	20.8	20.4	21.3	21.7	20.8	16.8	
		MCWBR	16.0	17.4	17.6	16.6	12.5	10.3	10.0	10.7	12.0	14.1	16.4	15.1	
Clear-Sky Solar Irradiance	taub		0.316	0.338	0.372	0.390	0.423	0.489	0.494	0.489	0.405	0.375	0.332	0.313	
	taud		2.394	2.310	2.301	2.297	2.234	2.084	2.083	2.083	2.294	2.381	2.465	2.449	
	Ebn at Noon		265	275	278	280	272	253	250	247	263	259	258	257	
	Edh at Noon		26	33	37	40	44	51	51	49	37	31	24	23	
All-Sky Solar Radiation	RadAvg		541	825	1125	1457	1649	1810	1840	1610	1313	905	624	449	
	RadStd		43	102	90	149	184	157	110	92	173	104	69	58	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (52 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+207	+101

Nomenclature: See separate page